D0. Choosing an analysis: start here

Answer four questions in order. The blue labels point to the focused diagrams D1 to D6.

1. What is the question?
Write it in one sentence first

- **Describe** — What do the data look like? → D1
- **Compare groups** — Do groups differ in the outcome? → D2 continuous, ordinal; D3 categorical
- **Assess an association** — Is X related to Y, possibly adjusted for others? → D3 both categorical; D4 otherwise
- **Predict** — Estimate Y for new people → D4 plus a train/test split
- **Discover groups** — Are there subgroups? No outcome variable → D5

2. What type is the outcome?
The variable you measure or explain

- **Continuous** — Measured on a scale e.g. SBP, BMI, CRP
- **Binary** — Two categories e.g. high BP yes/no
- **Nominal** — Unordered categories e.g. district
- **Ordinal** — Ordered categories e.g. satisfaction 1 to 5
- **Count** — Number of events e.g. clinic visits
- **Time-to-event** — Time until an event; some not yet observed (censored)

3. How were the observations collected?

- **Independent** — One measurement per person; people unrelated → D2 to D4
- **Paired** — Same people twice, or matched pairs → paired methods in D2, D3
- **Repeated** — Three or more measurements per person → D6 before analysing
- **Clustered** — People grouped in clinics, households, schools → D6 before analysing

4. How many groups or predictors?

- **Two groups** — e.g. women and men; t-test, Wilcoxon, 2 x 2 table
- **Three or more groups** — e.g. four districts; ANOVA, Kruskal-Wallis, r x c table
- **One predictor** — Correlation or simple regression
- **Several predictors** — Adjusted associations: multiple, logistic or Poisson regression

Use these diagrams to structure your thinking, not as automatic rules. Look at the data with plots, think about how they were collected, and report an effect estimate (difference, ratio, slope) with its 95% confidence interval, not only a p-value. Sample size alone does not decide between a t-test and a rank-based test, and neither does the p-value of a normality test.

D1. Describing data: summaries and plots by variable type

Always report n and the number of missing values. Plot the data before choosing a summary.

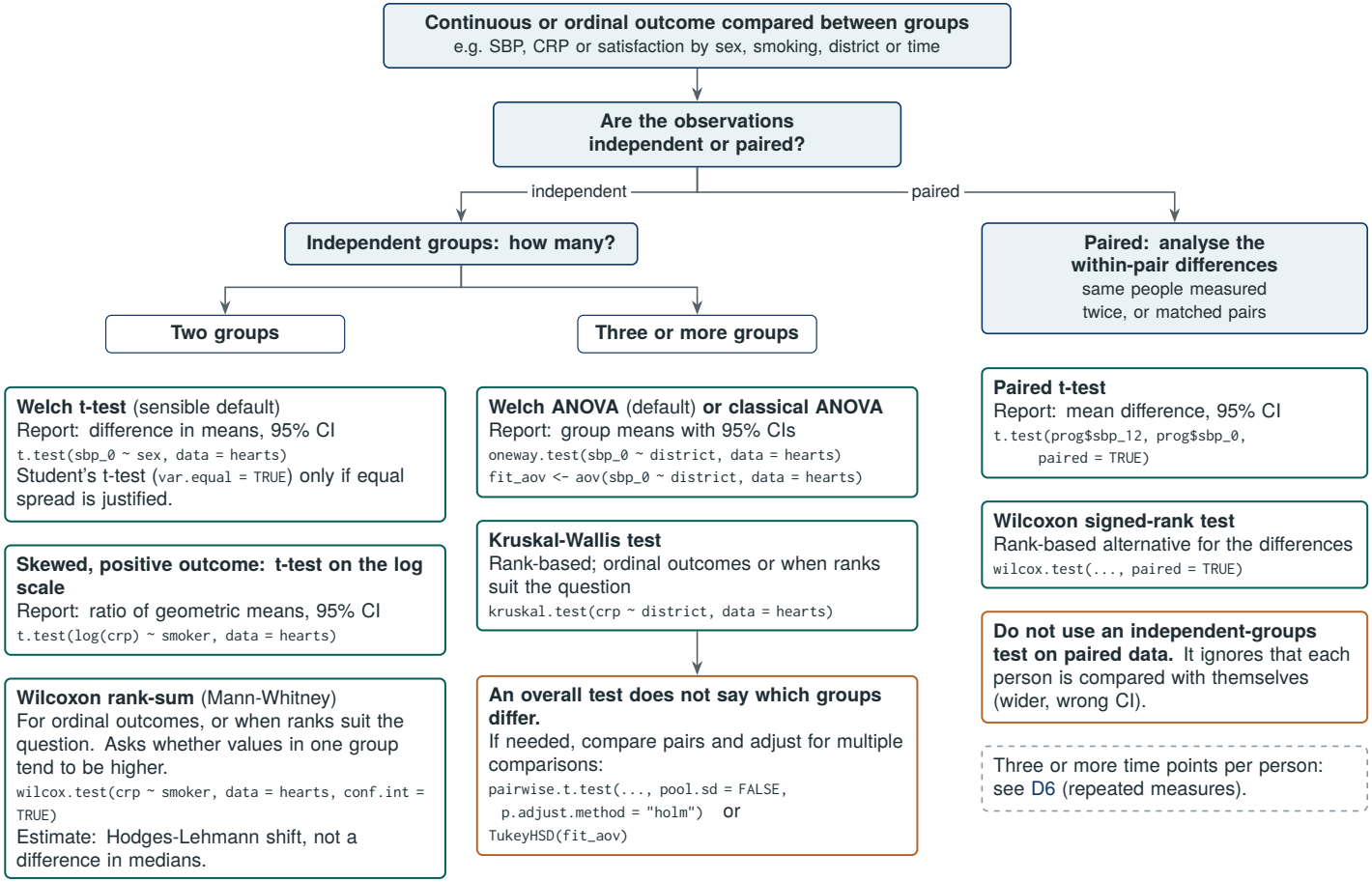
Variable type	Summarise with	Plot with	R, course data
One variable			
Binary or nominal e.g. smoker, district	Counts and percentages. State the denominator: everyone, or only those with a known value.	Bar chart of counts or percentages	`count(hearts, district)` `geom_bar()`
Ordinal e.g. satisfaction 1 to 5	Counts and percentages, kept in category order. Median and IQR can be added.	Bar chart in category order	`count(hearts, satisfaction)`
Continuous, roughly symmetric e.g. SBP	Mean and standard deviation (SD)	Histogram; boxplot	`mean(sbp_0); sd(sbp_0)` `geom_histogram(binwidth = 5)`

Continuous, skewed
e.g. CRP
	Median and interquartile range (IQR, 25th to 75th percentile). Mean and SD are pulled by extreme values.	Histogram, often on a log scale; boxplot	`median(crp, na.rm = TRUE)`
			`quantile(crp, c(0.25, 0.75), na.rm = TRUE)`
			`scale_x_log10()`

Count e.g. clinic visits	Mean and variance; a rate (events per person-time) when follow-up time differs	Bar chart of each count value	`sum(clinic_visits) / sum(followup_months)`
Two variables			
Continuous by group e.g. SBP by district	The summary above, given separately for each group, with n per group	Boxplots with the raw points	`group_by(district) |> summarise(...)` `geom_boxplot() + geom_jitter()`
Two categorical e.g. smoker by high BP	Contingency table with row percentages (denominator = each row total)	Bar chart of proportions	`tab <- table(hearts$smoker, hearts$high_bp_0)` `prop.table(tab, margin = 1)`
Two continuous e.g. age and SBP	Correlation, if the pattern is roughly linear or monotonic (see D4)	Scatterplot	`geom_point()`

D2. Comparing a continuous or ordinal outcome between groups

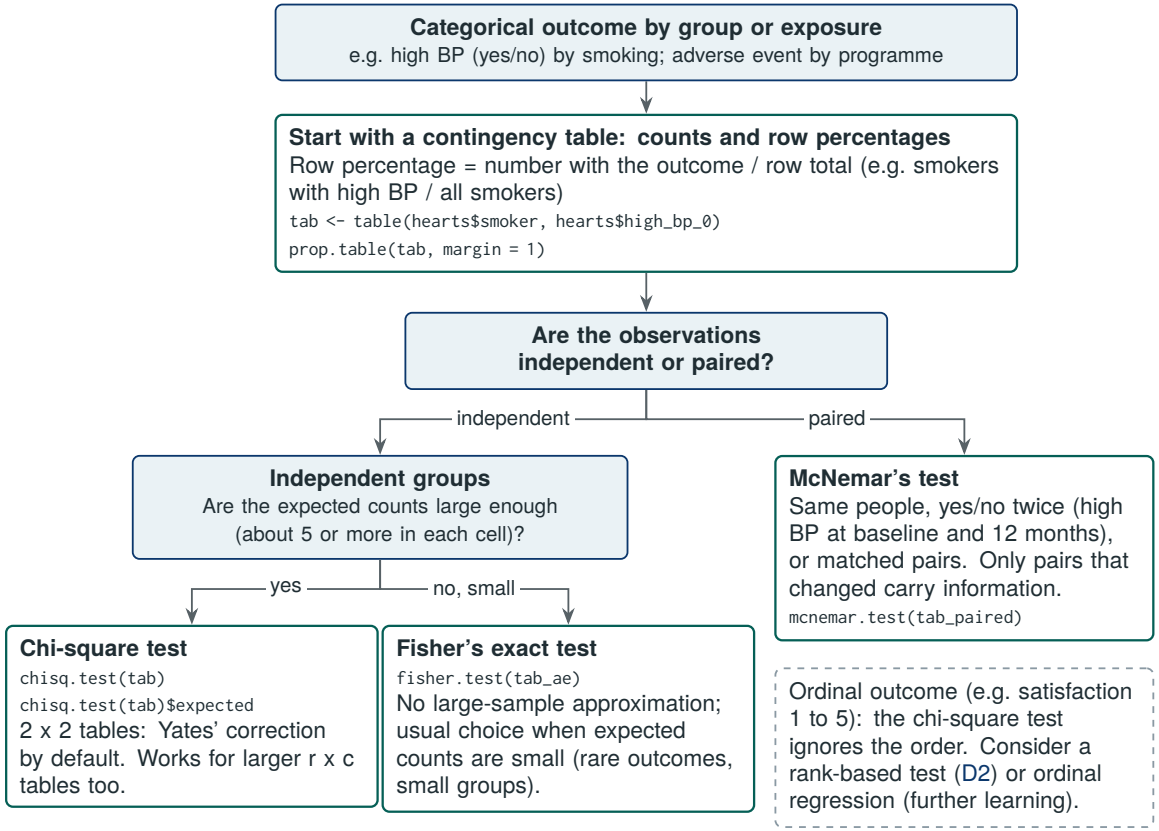
Report the size of the difference with a 95% CI. The test is only one part of the answer.



Assumptions and choices. t-tests and ANOVA assume independent observations (from the design) and approximately normal sample means. In large groups this holds even with moderate skew; with small groups, strong skew or outliers matter more. Check residual plots. Rank-based tests are not assumption-free: they also need independent observations, and they are not in general tests of medians. Do not choose a test from sample size alone or from a normality-test p-value: look at the data and decide which summary answers your question.

D3. Categorical outcomes: tables, tests and effect measures

Binary or nominal outcome compared between groups or exposures. Always state the denominator.



A significant test says the proportions differ somewhere. In tables larger than 2 x 2, look at the row percentages to see where. Report the effect measure (right) with its 95% CI, not only the p-value.

Which effect measure? It depends on the study design

Cohort study or randomised trial
People followed over time, so proportions are **risks**. Report risk difference (RD) and risk ratio (RR); OR also possible.
Trial example, adverse events: 14/240 (5.8%) with the programme vs 3/240 (1.2%) with standard advice. RR 4.67 (95% CI 1.36 to 16.03).

Cross-sectional study (one time point)
Proportions are **prevalences**. Report prevalence difference and prevalence ratio (PR); OR also possible.
Course example, high BP: 38/79 (48%) smokers vs 86/401 (21%) non-smokers. PR 2.24 (1.67 to 3.02); OR 3.39 (2.06 to 5.61).

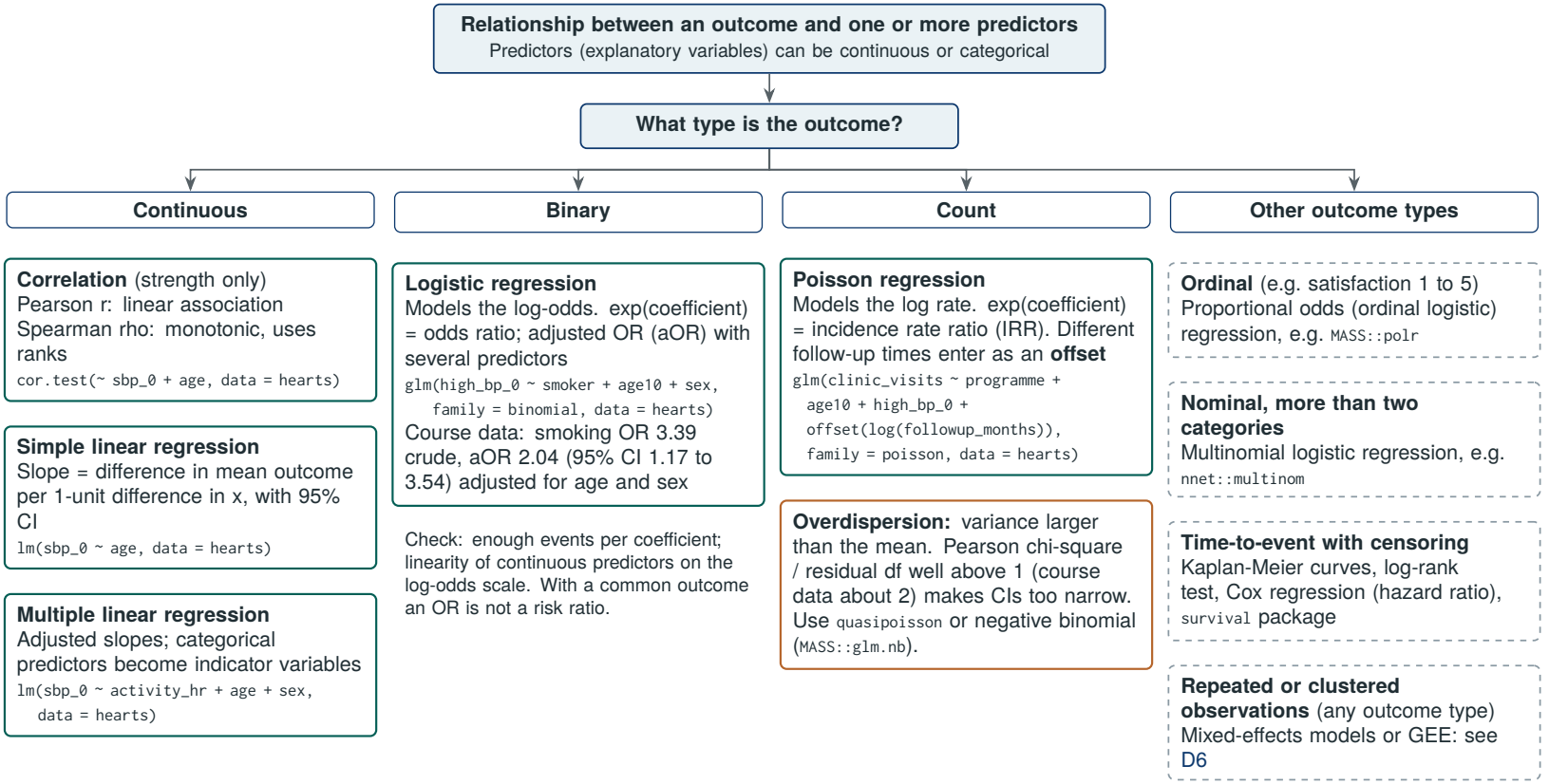
Case-control study
Cases and controls are chosen by outcome, so the proportion with the outcome is set by the design. Risks cannot be estimated: **only the OR** is interpretable.

OR is close to RR only when the outcome is rare (roughly under 10%). With a common outcome the OR is further from 1: do not read an OR of 3.4 as "3.4 times the risk".

Adjusting for a confounder
Stratify and combine: `mantelhaen.test(...)`, or logistic regression for an adjusted OR (aOR): see D4.

D4. Associations and regression models

Regression estimates how the outcome changes with each predictor, holding the other predictors constant. Dashed boxes are beyond this course: routes for further learning.



Explaining (estimating an association)
Choose adjustment variables from a causal diagram, not from p-values. Adjust for **confounders** (common causes of exposure and outcome), not for **mediators** (on the path from exposure to outcome). Association is not causation.

Predicting
Judge the model on data it has not seen: split into training and test sets (or cross-validate). Avoid **leakage**: no test-set or future information while building the model.

Checks for linear regression
Independence (from the design); linearity; constant spread; approximately normal residuals. Use `plot(fit)`: residuals vs fitted, Q-Q plot. The outcome itself need not be normal.

D5. Discovering groups: clustering

Unsupervised learning: there is no outcome. Clusters depend on the variables, the scaling, the method and k.

1. Purpose and variables

Write the question, e.g. "Are there distinct lifestyle profiles?"

Choose variables for substantive reasons. Every variable you add changes the distances.

Leave out outcomes you want to compare later (e.g. use activity, diet and BMI; keep SBP aside).

2. Prepare the data

Complete cases: `kmeans()` cannot handle missing values.

Very skewed variables may need a log transformation.

Variables on different scales or units?
e.g. minutes (0 to 900) and a score (0 to 10)

Yes (usual): standardise each variable to mean 0, SD 1

```
z <- scale(life[,
c("activity_min",
    "diet_score", "bmi")])
```

Without scaling, the variable with the largest numbers dominates.

3. Choose a method

k-means
Numeric variables; you set k in advance; finds compact, roughly round groups. Use several random starts.

```
set.seed(2024)
kmeans(z, centers = 3,
      nstart = 25)
```

Hierarchical clustering
Builds a tree (dendrogram) by merging the closest groups; cut it to get k groups. Choose a distance and a linkage (Ward's method is common).

```
hc <- hclust(dist(z),
     method = "ward.D2")
cutree(hc, k = 3)
```

4. Choose the number of clusters

Elbow plot: within-cluster sum of squares against k; look for a bend.

Average silhouette width: how well points sit in their own cluster (maximum 1).

Interpretability: can you describe and use the groups?

There is often no single right k. In the course data the elbow is gradual and all silhouettes are below 0.35: weak structure. Say so.

5. Check and describe

Describe clusters on the **original** scale (medians, means, n).

Check stability: other seeds, k, methods or subsets.

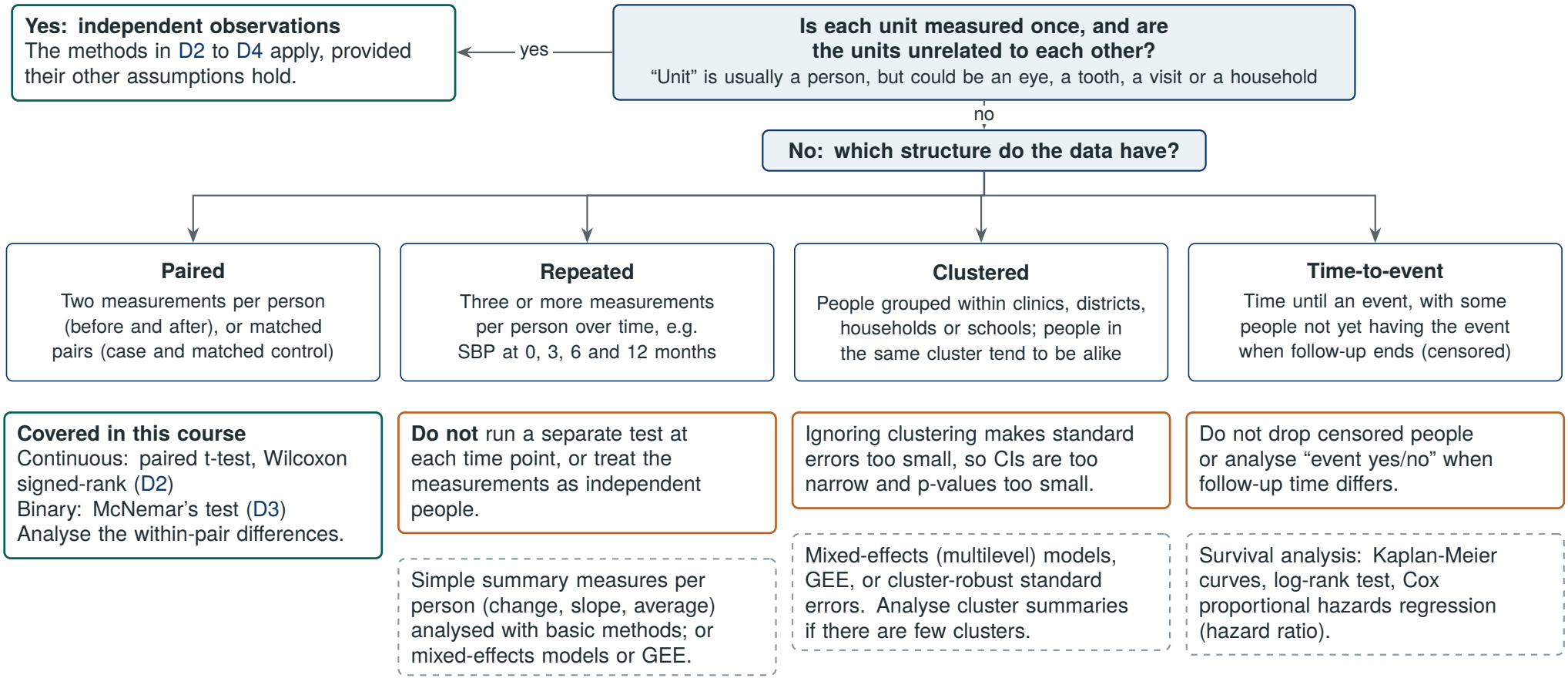
Compare clusters on variables **not** used to build them (e.g. age, SBP).

Apparent clusters are not automatically real. k-means always returns k groups, even from data with no groups. Do not test whether clusters differ on the variables used to create them.

Further learning: mixed numeric and categorical variables (Gower distance with `cluster::daisy` and `pam`); model-based clustering (Gaussian mixture models); principal component analysis for visualising many variables.

D6. Design check: are basic independent-observation methods appropriate?

The t-tests, chi-square tests and regression models in this course assume each observation is independent of the others.



Solid teal box: methods practised in the course. Orange-brown: common mistakes. Dashed grey: further learning beyond this course. A count over different follow-up times is handled in the course with a Poisson offset (D4).



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